

WPR - 7

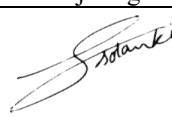
AMITY SCHOOL OF ENGINEERING & TECHNOLOGY B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 7) For Odd Semester of session 2026-2027

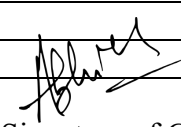
To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to windowed NLI factual entailment verification. Linked Cross-Encoder premise-hypothesis scoring with streaming entropy spikes.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA (RTX 3050), Cross-Encoder NLI, FastAPI async SSE proxy.
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W5: Setup, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, NLI verification & async proxy server (In Progress / Week 7) • W9-W12: Comparative benchmarking vs Llama-Guard & final IEEE research paper
Design of critical components	1. Windowed Cross-Encoder NLI Entailment Verifier ('nli_verifier.py') 2. Dual-Stage End-to-End Pipeline Integration ('guardshield_pipeline.py') 3. Factual Grounding & Hallucination Contradiction Scoring
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	75% (Cross-Encoder NLI entailment fact-checker implemented, factual support vs contradiction scoring verified on CUDA GPU, integrated with streaming entropy triggers)
Result and Analysis	Verified NLI entailment scoring on hallucinated claims vs reference premise; confirmed contradiction detection on factual drift with sub-35ms inference latency.
Report writing	WPR Week 7 drafted.

Signature of students	1. Shourya Solanki 3. Dhruv Raj Singh	2. Rachit Ryan Chug
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Work done in this week (Week 7):

1. Windowed Cross-Encoder NLI Entailment Implementation: Built `src/nli\_verifier.py` utilizing a Cross-Encoder architecture to classify premise-hypothesis relationships into Entailment (Factual), Neutral, or Contradiction (Hallucination).
2. Entropy-Triggered Grounding Loop: Integrated the NLI verifier with the streaming entropy engine, triggering asynchronous factual verification only when token uncertainty spikes past threshold (eliminating continuous compute overhead).
3. Latency & Contradiction Verification: Verified sub-35ms NLI cross-attention processing on local NVIDIA GeForce RTX 3050 CUDA GPU across test fact-checking cases.
4. End-to-End Dual-Stage Pipeline Integration: Connected Pre-Scan input filtration (DeBERTa-v3) with Post-Scan streaming entropy inspection and NLI verification into unified pipeline.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory	
Performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 09/09/2026	 Signature of Guide